

ARASH BABAMIRI

PhD Candidate - Building Energy Modeling

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WORK EXPERIENCE

Module Demonstrator / Teaching Assistant

- 📅 Sep 2024 - Present 📍 Dublin City University, Dublin, Ireland
- Facilitated laboratory sessions, tutorials, and practical coursework demonstrations for undergraduate engineering modules.

PhD Researcher - Building Energy Modelling

- 📅 Sep 2023 - Present 📍 Dublin City University, Dublin, Ireland
- Deliver Work Package 2 of the SEAI-funded ENABLE project, developing BIM-based energy models to assess residential retrofit performance across Irish housing archetypes.
 - Present research findings at international conferences, including CoBEE, EC3, and CIBSE.
 - Writing scientific publications (Publication list).

Researcher

- 📅 Jan 2021 - Sep 2023 📍 University of Kurdistan, Sanandaj, Iran
- Writing scientific articles on the energy and thermodynamics of integrated renewable systems.
 - Co-authored work on hybrid solar collector-chimney, thermoelectric and wind-turbine systems, and a micro-trigeneration cycle based on a Stirling engine, organic Rankine cycle, and adsorption chiller.

Mechanical Designer

- 📅 2021 - 2023 📍 DMO, Kashan, Iran
- Created detailed 3D models and technical drawings for diesel-generator canopies and chassis, and managed assembly processes.
 - Executed sheet-metal fabrication and laser-cutting projects while collaborating with cross-functional teams to streamline manufacturing workflows.
 - Automated the design process, reducing manual effort by up to 80%.

Researcher - Biomechanics (Freelance)

- 📅 Jan 2022 - Dec 2022 📍 Remote
- Collaborated with a research team on biomechanics projects, applying fluid mechanics and CFD analysis.
 - Co-authored three papers published in international conferences and journals on particle deposition in tracheobronchial airway models and dry powder inhaler aerosol deposition.

Industrial Designer

- 📅 2020 - 2021 📍 NTGC, Tehran, Iran
- Reverse-engineered industrial machine components for a glass manufacturer with more than 2,000 employees.
 - Developed detailed 3D models and industrial drawings.



PROFILE

Mechanical engineer and PhD candidate specialising in building energy modeling, with a background spanning thermodynamic analysis of heat recovery and renewable energy systems, BIM, and machine learning. Combines hands-on engineering design experience with coding, data analysis, surrogate modelling, and uncertainty quantification. Fast learner and diligent researcher with a strong record of peer-reviewed publications and conference presentations.

TECHNICAL SKILLS

- Building Performance & Energy Modelling:** EnergyPlus, IES VE, DesignBuilder, retrofit analysis, weather-data analysis, surrogate modelling, uncertainty quantification
- Irish Compliance & Sustainability:** Part L, NEAP, BER, NZEB, CIBSE TM52/TM54, overheating analysis, life-cycle assessment, One Click LCA
- Building Physics & HVAC:** HVAC design and system modelling, heating and cooling load calculations, thermodynamics, CFD with Ansys Fluent, daylight and sunlight analysis, thermal bridging
- BIM & Digital Coordination:** Revit MEP, Dynamo with Python, Navisworks, clash detection, ISO 19650, AutoCAD, SOLIDWORKS
- Programming & Data:** Python, machine learning, statistics, data analysis, workflow automation, XAI
- Technical Delivery:** Technical reporting and documentation, design reviews, ASHRAE 90.1 awareness, multidisciplinary coordination

SOFT SKILLS

- Written and verbal scientific communication
- Critical thinking and scientific data interpretation
- Interdisciplinary teamwork
- Cross-group coordination

WORK EXPERIENCE

Mechanical Design Engineer & Industrial Drafter

-  Feb 2019 - Nov 2021
-  Noritazeh, Tehran Province, Iran
- Designed new equipment and reverse-engineered broken components as part of the design team; followed up on parts manufacturing.
 - Deepened expertise in mechanical design and 3D CAD; translated design concepts into manufacturing- and construction-ready documentation.

Mechanical Designer / HVAC Designer Engineer

-  2018 - 2019
-  Pendar, Iran
- Designed and sized HVAC systems, including ductwork, piping, and equipment, for industrial buildings.
 - Coordinated with multidisciplinary teams to ensure MEP integration and compliance with ASHRAE 90.1 and local codes.
 - Prepared technical documentation including equipment specifications, BOQs, and vendor submittal reviews.

QC Manager

-  Jul 2017 - Apr 2018
-  Azmapayam Sarmad, Tehran Province, Iran
- Managed quality control processes, documentation, and technical reviews.

CERTIFICATIONS & TRAINING

- Green Building Concepts Foundations** - LinkedIn Learning, Feb 2026
- Nano Tips for Building Soft Skills with Elayne Fluker** - LinkedIn Learning, Jun 2025
- Communicating with Confidence** - LinkedIn Learning, Jun 2025
- Prompt Engineering: How to Talk to the AIs** - LinkedIn Learning, Jun 2025
- Practice Exam for LEED Green Associate** - LinkedIn Learning, Mar 2025
- Revit 2023: Essential Training for MEP** - LinkedIn Learning, Mar 2025
- Dynamo for Revit Python Scripting** - LinkedIn Learning, Feb 2025
- IES Virtual Environment Summer School** - IES Ltd.: Building Geometry Creation, Energy Analysis & Load Calculations, HVAC System Modelling, Daylight & Sunlight Modelling, Thermal Bridging & Parametric Simulations
- English Proficiency Certificate** - Duolingo English Test, May 2023
- GRE** - ETS, Sep 2021
- Applied Machine Learning in Python** - Coursera, Apr 2021
- Programming for Everybody (Getting Started with Python)** - Coursera, Mar 2021
- TOEFL** - ETS, Jul 2020

EDUCATION

PhD, Building Energy Modelling

-  Sep 2023 - Present
-  Dublin City University
- Designing and developing methods for modelling building energy consumption before and after retrofitting; publishing results in high-ranking scientific journals.

Master's Degree, Mechanical Engineering

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-  Malek Ashtar University of Technology
- Ranked 4th of 15 students.
 - Thesis: Stress and fatigue analysis of car chassis.

BSc, Mechanical Engineering

-  2011 - 2015
-  University of Kurdistan
- Ranked 2nd among 52 students.
 - Thesis: Energy and exergy analysis of a hybrid air conditioning system.

LEADERSHIP & PROFESSIONAL ACTIVITIES

- Organizing Committee** - IBPSA-Ireland National Student Symposium 2026, DCU: Co-organized a national symposium for the Irish building performance simulation community, including student presentations, poster sessions, and a Thesis-in-3-Minutes (3MT) competition.
- Presenter** - IBPSA Ireland seminar series: Presented PhD research and engaged with the Irish building performance simulation community.
- Speaker** - EC3 2026, Greece: Presented PhD research as part of the SEAI-funded ENABLE project.

RESEARCH INTERESTS

- Building energy modeling
- Energy efficiency and retrofit
- Uncertainty quantification
- Surrogate modelling
- BIM and computational workflows
- Machine learning
- Renewable energy systems